

# Renewable Energies (2026-27)

Project Statement: Pre-FEED of an Autonomous Island Hybrid Energy System

## 1 Context & Design Objective

Your engineering consultancy has been hired by the Island Energy Council to perform a Pre-Front-End Engineering Design (Pre-FEED) study to transition your **assigned isolated island** (see [Appendix A](#)) away from its current baseline heavy fuel oil (HFO) thermal power plant towards energy autonomy.

**Objective:** As a project team of 4 engineering consultants, you must evaluate local renewable resources and propose an optimized hybrid system (Wind, Solar, and Hydro) that matches your assigned island's electricity consumption pattern. Your goal is to maximize Variable Renewable Energy (VRE) penetration while defining the absolute minimum dispatch required from the baseline thermal plant to ensure 100% grid demand coverage. Python scripting and AI-assisted tools for data extraction, dynamic time-series dispatch, and multi-variable optimization are strongly encouraged.

## 2 Island Demand Profile & Grid Constraints

To keep computational efforts aligned with the project scope, the annual demand is compressed into **12 Representative Days** (one typical 24-hour profile for each month). You are provided with a dimensionless multiplier matrix file, `Universal_Load_288.csv`, representing the normalized shape of the island's demand ( $\mathbf{M}_{0-1}$ ).

### 2.1 Grid Sizing Parameters

Each project group will be assigned one of the four island archetypes defined in [Appendix A](#). Apply the following absolute power system thresholds from [Table 4](#) to scale the dimensionless matrix for your assigned island:

- **Base Load** ( $P_{\text{base}}$ ): Refer to [Table 4](#) [MW]
- **Peak Load** ( $P_{\text{peak}}$ ): Refer to [Table 4](#) [MW]

### 2.2 Mathematical Reconstruction

Reconstruct the absolute hourly electrical power demand ( $P_{\text{demand}}$  [MW]) for any given month ( $m$ ) and hour ( $h$ ) using the following linear transformation:

$$P_{\text{demand}}(m, h) = P_{\text{base}} + \mathbf{M}_{0-1}(m, h) \times (P_{\text{peak}} - P_{\text{base}}) \quad (1)$$

To calculate the final annual energy volume ( $E_{\text{annual}}$  [MWh]) for your economic models, integrate the scaled hours using the standard number of days per month ( $N_m$ ):

$$E_{\text{annual}} = \sum_{m=1}^{12} \left[ \left( \sum_{h=0}^{23} P_{\text{demand}}(m, h) \times 1 \text{ hour} \right) \times N_m \right] \quad (2)$$

### 3 Renewable Resource Sourcing Parameters

**Important Data Processing Note:** While environmental data platforms (such as PVGIS, Renewables.ninja, or ERA5) typically export full 8760-hour (365 days) datasets, you **must compress this data** to align with the island’s demand profile constraint. For both wind and solar resources, compute the monthly average for each hour of the day to generate a **12 Representative Days** matrix ( $12 \times 24$  hours = 288 hours total). All subsequent power generation and dispatch calculations should be performed on this reduced 288-hour timeline to ensure consistent temporal alignment with the load profile.

#### 3.1 A. Wind Resource

To model wind boundary layer physics, site climatology, aerodynamic limits, and turbine yield:

- **Site Selection & IEC Turbine Class Matching:** Using topographical tools (e.g., **Global Wind Atlas**), select an optimal onshore location. Extract the site’s mean wind speed ( $\bar{U}$ ) and Weibull distribution parameters (scale parameter  $c$ , shape parameter  $k$ ) to determine the appropriate **IEC Wind Class** (Class I, II, or III) and justify your chosen commercial wind turbine model.
- **Terrain Roughness Length ( $z_0$ ):** Estimate the surface roughness parameter  $z_0$  based on visual inspection of the terrain and standard wind engineering tables.
- **Velocity Extrapolation & Data Compression:** Extract reference hourly wind speed time-series (e.g., via **Renewables.ninja**, NASA POWER, or ERA5), scale them to the selected turbine hub height using the Logarithmic Wind Profile, and compress the profile into the 288-hour ( $12 \times 24$  h) representative matrix for grid dispatch modeling.
- **Aerodynamic & Power Curve Mapping:** Pass hub-height wind speeds through the selected turbine’s commercial power curve to calculate hourly power generation while evaluating operational output against physical boundaries (Betz Limit:  $C_{P,\text{max}} = 59.3\%$ ).

#### 3.2 B. Solar Resource & Yield Modeling

To perform the solar geometry, resource assessment, and hourly yield modeling for the Pre-FEED study:

- **Site Selection & Footprint:** Utilizing topographical tools and the **Global Solar Atlas**, select and justify an optimal footprint on the assigned island for a utility-scale PV plant, accounting for land area requirements (ha/MW<sub>p</sub>).
- **Time Synchronization:** Convert UTC or Standard Time vectors into Legal Clock Time ( $LCT$ ) to properly align solar production with local hourly demand profiles. To achieve this within the compressed 12-representative-day framework, assume the **15th day of each month** as your reference date to calculate the Equation of Time ( $E_t$ ), and combine it with the constant site longitude correction ( $L_{\text{std}} - L_{\text{loc}}$ ).

- **Irradiance Sourcing & Transposition:** Extract typical hourly irradiance ( $GHI$ ,  $DHI$ ,  $DNI$ ) and ambient temperature ( $T_a$ ) data via **PVGIS** or **Renewables.ninja**, compressing it into the 288-hour matrix (12 representative days). Transpose horizontal irradiance to Plane of Array ( $G_{POA}$ ), optimizing fixed tilt ( $\beta$ ) and azimuth ( $\rho$ ) angles for maximum annual capture.
- **Hourly Power Calculation:** Model hourly AC power output ( $P_{PV}$ ) using the **Fast-Estimate 1-Parameter Model**:
  1. Compute hourly cell operating temperature ( $T_c$ ) using the NOCT thermal model driven by  $T_a$  and  $G_{POA}$ .
  2. Derive net AC generation using commercial module parameters ( $P_{DC}$ ,  $G_{STC}$ ,  $\gamma_p$ , NOCT) and Balance of System losses ( $PR_{BOS}$ ).

### 3.3 C. Hydropower Resource & Yield Modeling

Using synthetic hydrological data for a prospective run-of-river site on your assigned island:

- **Head Definitions & Dynamic Loss Modeling:** Extract Gross Head ( $H_b$ ) from Table 4 [m]. Model penstock friction loss dynamically as a function of operational flow  $Q$  relative to your turbine design discharge  $Q_d$ , assuming a design friction loss coefficient of 8% at nominal flow:

$$H_n(Q) = H_b \times \left[ 1 - 0.08 \times \left( \frac{Q}{Q_d} \right)^2 \right] \quad (3)$$

Calculate hourly Net Head ( $H_n$ ) for all usable flow conditions ( $Q \leq Q_d$ ).

- **Environmental Flow & Usable Flow:** Deduct exactly **15%** of the streamflow as mandatory environmental flow. Determine the net usable flow available for power generation, ensuring it is properly bounded by the plant's rated design discharge.
- **Flow Duration Curve (FDC) & Turbine Selection:** Construct the empirical FDC using streamflow data from Table 5. Evaluate the non-dimensional specific speed at a valid 50 Hz grid-synchronized rotational speed. Select and justify an appropriate turbine topology (Pelton, Francis, or Kaplan) using e.g., the Cordier diagram.
- **Power Output Calculation:** Calculate the hourly net electrical power output delivered to the grid, accounting for variable net head, overall electrical efficiency, and turbine part-load efficiency.

**Note on Hydropower Siting:** The hydrological profiles and gross head values provided in Table 4 and Table 5 represent prospective, synthetic run-of-river sites for this Pre-FEED study. Students must model this asset strictly as a run-of-river facility and should not attempt to replicate existing real-world island infrastructure (such as pumped-hydro plants).

## 4 Baseline Thermal Power Plant & Economic Framework

Evaluate your hybrid design against the thermal baseline using Levelized Cost of Energy (LCOE) calculations over the project lifecycle.

#### 4.1 Financial & Baseline Thermal Inputs

To ensure physical consistency and cross-group comparability, the baseline HFO thermal power plant parameters and thermo-chemical fuel properties are standardized in Table 1.

Table 1: Baseline Thermal Plant Thermodynamic and Fuel Parameters

| Parameter                   | Symbol      | Standardized Value                       |
|-----------------------------|-------------|------------------------------------------|
| Nominal Thermal Efficiency  | $\eta_{th}$ | 36%                                      |
| HFO Lower Heating Value     | $LHV$       | 40.5 MJ/kg (11.25 kWh <sub>th</sub> /kg) |
| HFO Fuel Density            | $\rho$      | 0.95 kg/L                                |
| Fuel Carbon Emission Factor | $EF_{HFO}$  | 3.15 kg CO <sub>2</sub> /kg HFO          |

Students must research, justify, and document the following market and policy parameters:

- **Project Lifespan ( $t$ ):** Fixed at 20 years.
- **Baseline Discount Rate ( $r$ ):** Define and justify a discount rate (%) based on island project financing risk.
- **HFO Fuel Delivered Cost:** Research and justify the baseline heavy fuel oil cost (€/L) delivered to your assigned island.
- **Carbon Tax Policy:** Define and justify a municipal/regional carbon tax (€/tCO<sub>2</sub>) applied to the baseline thermal generation.

#### 4.2 Asset Cost Boundaries & Sensitivities

Utilize CAPEX, Fixed OPEX, and asset degradation rates within the following engineering thresholds (Table 2). Incorporate annual asset degradation rates into your 20-year LCOE calculations and conduct a 2-way sensitivity analysis varying discount rate ( $r$ ) against baseline HFO fuel price (€/L) to identify the economic tipping point for renewable integration. Ensure your analysis clearly compares the LCOE of the new integrated solution against the baseline LCOE of the original 100% HFO plant.

**Thermal Plant Sizing & CAPEX Rule:** The island’s electricity supply originally relies 100% on the existing HFO thermal plant. Therefore, you must assume the baseline thermal plant has an installed capacity equal to your assigned island’s Peak Load ( $P_{peak}$ ). By default, treat this baseline plant’s CAPEX as €0 (sunk asset). However, the existing legacy plant is inefficient at low part-loads (< 30%). If you retain it as a €0 sunk asset, you may be forced to curtail (waste) variable renewable energy during high-generation hours just to keep the thermal plant running above its minimum 30% threshold. If you choose to avoid this curtailment by replacing the legacy plant with new, smaller, fast-response peaker units to balance VRE fluctuations, you must apply the Thermal Plant CAPEX penalty to your newly sized thermal capacity.

#### 4.3 Grid Resilience & Backup LCOE Comparison

The 288-hour compressed profile inherently smoothes out multi-day renewable droughts (e.g., 10 consecutive hours of near-zero wind and solar generation). To evaluate real-world grid security, calculate and compare two distinct lifecycle cost scenarios:

- **Scenario A (Optimized Hybrid):** The baseline LCOE of your proposed system, using your dynamically downsized thermal plant capacity.

Table 2: Sizing and Financial Thresholds

| Asset Type          | CAPEX Range<br>(€/kW) | Fixed OPEX<br>(%/year) | Degradation Rate<br>(%/year) |
|---------------------|-----------------------|------------------------|------------------------------|
| Onshore Wind        | €1,300 – €1,600       | 2.5%                   | 1.0%                         |
| Utility Solar PV    | €800 – €1,100         | 1.8%                   | 0.5%                         |
| Run-of-River Hydro  | €2,500 – €3,500       | 3.0%                   | 0.2%                         |
| Thermal Plant (HFO) | €1,000 – €1,400*      | 4.0%                   | 0.3%                         |

\* For the thermal plant: CAPEX is €0 if retaining legacy plant; variable OPEX (fuel) must be calculated dynamically. For renewables, variable OPEX can be neglected.

- **Scenario B (100% Firm Security Backup):** The LCOE of your proposed system assuming the island must retain or replace a full 100% peak load ( $P_{\text{peak}}$ ) thermal backup plant (incorporating the full thermal CAPEX penalty if new peaker units are required) as cold-standby reserve to guarantee  $N - 1$  reliability.

Quantify the cost difference  $\Delta\text{LCOE}$  (€/MWh) between Scenario A and Scenario B to evaluate the “Reliability Premium” required to protect the island against extreme sub-hourly and multi-day VRE drop-offs.

## 5 Deliverables & Evaluation

This project represents **30% of your final grade**. The evaluation models a real-world consulting proposal pitch to a municipal client:

1. **Pre-FEED Executive Form Report:** Submitted at the project delivery deadline, each group must fill out and submit the provided standardized `Report_Template_RE_2026-27.tex` form in PDF format. This report serves to provide a comprehensive overview of the work performed, capturing your resource calculations (log wind profiles, POA adjustments, FDCs), hourly  $12 \times 24$  h dispatch balance plots, final capacity sizing, the LCOE sensitivity matrix, and the Scenario A vs. Scenario B Reliability Premium comparison.
2. **Consulting Pitch Presentation:** Taking place after the report delivery deadline, this 20-minute meeting simulates a presentation to the Island Energy Council. The presentation content must be strictly aligned with your submitted report and serves to clarify project details as well as evaluate individual student knowledge:
  - **10-Minute Pitch:** Deliver a commercial and technical pitch convincing the Council to invest in your hybrid energy system layout over the thermal baseline, consistent with the findings in your submitted report.
  - **10-Minute Q&A:** Defend your technical sizing, dynamic dispatch strategies, aerodynamic assumptions, and economic modeling under direct client questioning. This session will be used to clarify report details and assess each student’s individual understanding.

## A Island Datasets & Hydrological Parameters

To ensure a diverse analysis of isolated grid topologies, each project team is assigned a specific island archetype from either the Azores or Canary archipelagos. Table 3 outlines the specific island assigned to your group number. All subsequent environmental data extraction, grid sizing, and geographical siting must be conducted exclusively for your assigned location.

Table 3: Project Group Island Assignments

| Group Number    | Assigned Island      |
|-----------------|----------------------|
| Groups 1 to 4   | Flores (Azores)      |
| Groups 5 to 8   | São Jorge (Azores)   |
| Groups 9 to 12  | El Hierro (Canaries) |
| Groups 13 to 17 | La Palma (Canaries)  |

Table 4 defines the power grid thresholds and hydraulic head for each of the four island archetypes. Table 5 presents the synthetic monthly average river flow rates. The environmental bypass flow rate is fixed at **15%** for all islands.

Table 4: Island Grid Sizing Constraints and Hydrological Parameters

| Archipelago     | Island Name | Base Load ( $P_{\text{base}}$ ) | Peak Load ( $P_{\text{peak}}$ ) | Gross Head ( $H_b$ ) |
|-----------------|-------------|---------------------------------|---------------------------------|----------------------|
| <b>Azores</b>   | Flores      | 1.2 MW                          | 3.8 MW                          | 120 m                |
| <b>Azores</b>   | São Jorge   | 1.5 MW                          | 4.5 MW                          | 150 m                |
| <b>Canaries</b> | El Hierro   | 3.5 MW                          | 8.2 MW                          | 150 m                |
| <b>Canaries</b> | La Palma    | 4.0 MW                          | 11.5 MW                         | 200 m                |

Table 5: Synthetic Monthly Average River Flow Rates ( $m^3/s$ )

| Island           | Jan  | Feb  | Mar  | Apr  | May  | Jun  | Jul  | Aug  | Sep  | Oct  | Nov  | Dec  |
|------------------|------|------|------|------|------|------|------|------|------|------|------|------|
| <b>Flores</b>    | 0.85 | 0.90 | 0.75 | 0.55 | 0.35 | 0.20 | 0.12 | 0.10 | 0.18 | 0.35 | 0.60 | 0.80 |
| <b>São Jorge</b> | 0.60 | 0.65 | 0.50 | 0.40 | 0.25 | 0.15 | 0.10 | 0.08 | 0.15 | 0.30 | 0.45 | 0.55 |
| <b>El Hierro</b> | 0.50 | 0.55 | 0.40 | 0.28 | 0.18 | 0.10 | 0.05 | 0.04 | 0.08 | 0.20 | 0.35 | 0.48 |
| <b>La Palma</b>  | 0.70 | 0.75 | 0.60 | 0.45 | 0.30 | 0.20 | 0.12 | 0.10 | 0.18 | 0.35 | 0.50 | 0.65 |